

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Augmentation of MAOIs has not been systematically studied, and this is something for the expert, to be done with caution and with careful monitoring, but may be somewhat less risky with moclobemide than with other MAOIs
- * A stimulant such as d-amphetamine or methylphenidate (with caution; may activate bipolar disorder and suicidal ideation)
- Lithium
- Mood-stabilizing anticonvulsants
- Atypical antipsychotics (with special caution for those agents with monoamine reuptake blocking properties, such as lurasidone, ziprasidone, and zotepine)

Tests

- Patients should be monitored for changes in blood pressure

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in monoamines in parts of the brain and body and at receptors other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of norepinephrine on vascular smooth muscle causing hypotension)
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- Insomnia, dizziness, agitation, anxiety, restlessness
- Galactorrhea
- Rare hypertension

Life-Threatening or Dangerous Side Effects

- Hypertensive crisis (especially when MAOIs are used with certain tyramine containing foods –much reduced risk compared to irreversible MAOIs)
- Induction of mania in patients with bipolar disorder
- Rare seizures

Weight Gain



- Reported but not expected

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Switch to an SSRI or newer antidepressant

Best Augmenting Agents for Side Effects

- Trazodone (with caution) for insomnia
- Benzodiazepines for insomnia

Dosing and Use

Usual Dosage Range

- 300–600 mg/day

Dosage Forms

- Tablet 100 mg scored, 150 mg scored

How to Dose

- Initial 300 mg/day in 3 divided doses after a meal; increase dose gradually; maximum dose generally 600 mg/day; minimum dose generally 150 mg/day



Dosing Tips

- * At higher doses, moclobemide also inhibits MAO-B and thereby loses its selectivity for MAO-A, with uncertain clinical consequences
- * Taking moclobemide after meals as opposed to before may minimize the chances of interactions with tyramine
- May be less toxic in overdose than TCAs and older MAOIs
- Clinical duration of action may be longer than biological half-life and allow twice daily dosing in some patients, or even once daily dosing, especially at lower doses

Overdose

- Agitation, aggression, behavioral disturbances, gastrointestinal irritation

Long-Term Use

- MAOIs may lose efficacy long-term

Habit Forming

- Dependence to MAOIs reported but rare

How to Stop

- Taper not generally necessary

Pharmacokinetics

- Partially metabolized by CYP450 2C19 and 2D6
- Inactive metabolites

- Elimination half-life approximately 1–4 hours
- Clinical duration of action at least 24 hours

Drug Interactions

- Tramadol (serotonin reuptake inhibitor) may increase the risk of seizures in patients taking an MAOI
- Can cause a fatal “serotonin syndrome” when combined with drugs that block serotonin reuptake, so do not use with a serotonin reuptake inhibitor or for 5 half-lives after stopping the serotonin reuptake inhibitor (see [Table 1](#) after Pearls)
- Hypertensive crisis with headache, intracranial bleeding, and death may result from combining MAOIs with sympathomimetic drugs (e.g., amphetamines, methylphenidate, cocaine, dopamine, epinephrine, norepinephrine)
- Do not combine with another MAOI, alcohol, or guanethidine
- Adverse drug reactions can result from combining MAOIs with tricyclic/tetracyclic antidepressants and related compounds, including carbamazepine, cyclobenzaprine, and mirtazapine, and should be avoided except by experts
- MAOIs in combination with spinal anesthesia may cause combined hypotensive effects
- Combination of MAOIs and CNS depressants may enhance sedation and hypotension
- Cimetidine may increase plasma concentrations of moclobemide
- Moclobemide may enhance the effects of NSAIDs such as ibuprofen
- Risk of hypertensive crisis may be increased if moclobemide is used concurrently with levodopa or other dopaminergic agents

Table 1. Drugs contraindicated due to risk of serotonin syndrome/toxicity

Do Not Use:			
Antidepressants	Drugs of Abuse	Opioids	Other
SSRIs	MDMA (ecstasy)	Meperidine	Non-subcutaneous sumatriptan
SNRIs	Cocaine	Tramadol	Chlorpheniramine
Clomipramine	Methamphetamine	Methadone	Brompheniramine
Imipramine	St. John’s wort	High-dose or injected amphetamine	Dextromethorphan Lumateperone Ziprasidone

Other Warnings/Precautions

- Use still requires a low-tyramine diet, although more tyramine may be tolerated with moclobemide than with other MAOIs before eliciting a hypertensive reaction (see [Table 2](#) after Pearls)
- Patient and prescriber must be vigilant to potential interactions with any drug, including antihypertensives and over-the-counter cough/cold preparations
- Over-the-counter medications to avoid include cough and cold preparations, including those containing dextromethorphan (serotonin reuptake inhibitor), nasal decongestants (tablets, drops, or spray), hay-fever medications, sinus medications, asthma inhalant medications, anti-appetite medications, weight reducing preparations, “pep” pills (see [Table 3](#) after Pearls)
- Use cautiously in hypertensive patients
- Moclobemide is not recommended for use in patients who cannot be monitored closely

Table 2. Dietary guidelines for patients taking MAOIs*

Foods to avoid**	Foods allowed
Dried, aged, smoked, fermented, spoiled, or improperly stored meat, poultry, and fish	Fresh or processed meat, poultry, and fish; properly stored pickled or smoked fish
Broad bean pods	All other vegetables
Aged cheeses	Processed cheese slices, cottage cheese, ricotta cheese, yogurt, cream cheese
Tap and unpasteurized beer	Canned or bottled beer and alcohol
Marmite	Brewer’s and baker’s yeast
Banana peel	Bananas, avocados, raspberries
Sauerkraut, kimchee	Peanuts
Soy products/tofu	
Tyramine-containing nutritional supplement	

* Consult up-to-date listing of tyramine content, as most foods now have low tyramine amounts

** Not necessary for 6 mg transdermal or low-dose oral selegiline

Table 3. Drugs that boost norepinephrine: should only be used with caution with MAOIs

Use With Caution:

Decongestants	Stimulants	Antidepressants with norepinephrine reuptake inhibition	Other
Phenylephrine	Amphetamines	Most tricyclics	Phenylephrine
Pseudoephedrine	Methylphenidate Cocaine Methamphetamine Modafinil Armodafinil	NRIs NDRIs	Local anesthetics containing vasoconstrictors

Do Not Use

- If patient is taking meperidine (pethidine) (serotonin reuptake inhibitor)
- If patient is taking a sympathomimetic agent or taking guanethidine
- If patient is taking another MAOI
- If patient is taking any agent that can inhibit serotonin reuptake (e.g., SSRIs, tramadol, milnacipran, duloxetine, venlafaxine, clomipramine, imipramine, etc.)
- If patient is in an acute confusional state
- If patient has pheochromocytoma or thyrotoxicosis
- If there is a proven allergy to moclobemide

Special Populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Plasma concentrations are increased
- May require one-half to one-third of usual adult dose

Cardiac Impairment

- Cardiac impairment may require lower than usual adult dose
- Patients with angina pectoris or coronary artery disease should limit their exertion

Elderly

- Elderly patients may have greater sensitivity to adverse effects, but many tolerate MAOIs

- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older

Children and Adolescents

- Not recommended for use in children under age 18
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients

Pregnancy

- Not generally recommended for use during pregnancy, especially during first trimester
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio

Breast Feeding

- Some drug is found in mother's breast milk
- Effects on infant are unknown
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Should evaluate patient for treatment with an antidepressant with a better risk/benefit ratio

The Art of Psychopharmacology

Potential Advantages

- Atypical depression
- Treatment-resistant depression or anxiety disorders

Potential Disadvantages

- Patients noncompliant with dietary restrictions, concomitant drug restrictions, and twice daily dosing after meals

Primary Target Symptoms

- Depressed mood

Pearls

- MAOIs are generally reserved for third- or fourth-line use after SSRIs, SNRIs, and combinations of newer antidepressants have failed
- Patient should be advised not to take any prescription or over-the-counter drugs without consulting his/her doctor because of possible drug interactions with the MAOI
- Headache is often the first symptom of hypertensive crisis
- Moclobemide has a much reduced risk of interactions with tyramine than nonselective MAOIs
- Especially at higher doses of moclobemide, foods with high tyramine need to be avoided (see [Table 2](#))
- The rigid dietary restrictions may reduce compliance
- * May be a safer alternative to classical irreversible nonselective MAO-A and MAO-B inhibitors with less propensity for tyramine and drug interactions and hepatotoxicity (although not entirely free of interactions)
- May not be as effective at low doses, and may have more side effects at higher doses
- Moclobemide's profile at higher doses may be more similar to classical MAOIs
- * Myths about the danger of dietary tyramine can be exaggerated, but prohibitions against concomitant drugs often not followed closely enough
- Orthostatic hypotension, insomnia, and sexual dysfunction are often the most troublesome common side effects
- * MAOIs should be for the expert, especially if combining with agents of potential risk (e.g., stimulants, trazodone, TCAs)
- * MAOIs should not be neglected as therapeutic agents for the treatment-resistant
- For treatment-resistant patients, experts can give a tricyclic/tetracyclic antidepressant other than clomipramine or imipramine simultaneously with an MAOI for patients who fail to respond to numerous other antidepressants
- Use of MAOIs with clomipramine is always prohibited because of the risk of serotonin syndrome and death due to serotonin reuptake inhibition
- Amoxapine may be a preferred tricyclic/tetracyclic antidepressant to combine with an MAOI in heroic cases due to its theoretically protective 5HT_{2A} antagonist properties
- If this option is elected, start the MAOI with the tricyclic/tetracyclic antidepressant simultaneously at low doses after appropriate drug washout, then alternately increase doses of these agents every few days to a week as tolerated
- Although very strict dietary and concomitant drug restrictions must be observed to prevent hypertensive crises and serotonin syndrome, the most common side effects of MAOI and tricyclic/tetracyclic combinations may be weight gain and orthostatic hypotension

Suggested Reading

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Modafinil

Therapeutics

Brands

- Provigil
- Alertec
- Modiodal
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine reuptake inhibitor (D-RI)
- Wake-promoting

Commonly Prescribed for

(bold for FDA approved)

- **Reducing excessive sleepiness in patients with narcolepsy and shift work sleep disorder**

- **Reducing excessive sleepiness in patients with obstructive sleep apnea/hypopnea syndrome (OSAHS) (adjunct to standard treatment for underlying airway obstruction)**
- Attention deficit hyperactivity disorder (ADHD)
- Fatigue and sleepiness in depression
- Fatigue in multiple sclerosis
- Bipolar depression

How the Drug Works

- Unknown, but clearly different from classical stimulants such as methylphenidate and amphetamine
- Binds to and requires the presence of the dopamine transporter; also requires the presence of alpha adrenergic receptors
- Hypothetically acts as an inhibitor of the dopamine transporter
- Increases neuronal activity selectively in the hypothalamus
- * Presumably enhances activity in hypothalamic wakefulness center (TMN, tuberomammillary nucleus) within the hypothalamic sleep-wake switch by an unknown mechanism
- * Activates TMN that release histamine
- * Activates other hypothalamic neurons that release orexin/hypocretin

How Long Until It Works

- Can immediately reduce daytime sleepiness and improve cognitive task performance within 2 hours of first dosing
- Can take several days to optimize dosing and clinical improvement

If It Works

- * Improves daytime sleepiness and may improve attention as well as fatigue
- * Does not generally prevent one from falling asleep when needed
- May not completely normalize wakefulness
- Treat until improvement stabilizes and then continue treatment indefinitely as long as improvement persists (studies support at least 12 weeks of treatment)

If It Doesn't Work

- * Change dose; some patients do better with an increased dose but some actually do better with a decreased dose
- Augment or consider an alternative treatment for daytime sleepiness, fatigue, or ADHD



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Modafinil is itself an adjunct to standard treatments for obstructive sleep apnea/hypopnea syndrome (OSAHS); if continuous positive airway pressure (CPAP) is the treatment of choice, a maximal effort to

treat first with CPAP should be made prior to initiating modafinil and CPAP should be continued after initiation of modafinil

* Modafinil is itself an augmenting therapy to antidepressants for residual sleepiness and fatigue in major depressive disorder

- Best to attempt another monotherapy prior to augmenting with other drugs in the treatment of sleepiness associated with sleep disorders or problems concentrating in ADHD
- Combination of modafinil with stimulants such as methylphenidate or amphetamine or with atomoxetine for ADHD has not been systematically studied
- However, such combinations may be useful options for experts, with close monitoring, when numerous monotherapies for sleepiness or ADHD have failed

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Unknown
- CNS side effects presumably due to excessive CNS actions on various neurotransmitter systems

Notable Side Effects

- * Headache (dose-dependent)
- Anxiety, nervousness, insomnia
- Dry mouth, diarrhea, nausea, anorexia
- Pharyngitis, rhinitis, infection
- Hypertension
- Palpitations



Life-Threatening or Dangerous Side Effects

- Transient ECG ischemic changes in patients with mitral valve prolapse or left ventricular hypertrophy have been reported (rare)
- Rare activation of (hypo)mania, anxiety, hallucinations, or suicidal ideation
- Rare severe dermatologic reactions (Stevens-Johnson syndrome and others)
- Angioedema, anaphylactoid reactions, and multi-organ hypersensitivity reactions have been reported

Weight Gain



- Reported but not expected

Sedation



- Reported but not expected
- Patients are usually awakened and some may be activated

What to Do About Side Effects

- Wait
- Lower the dose
- Give only once daily
- Give smaller split doses 2 or more times daily
- For activation or insomnia, do not give in the evening
- If unacceptable side effects persist, discontinue use
- For life-threatening or dangerous side effects, discontinue immediately (e.g., at first sign of a drug-related rash)

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 200 mg/day in the morning

Dosage Forms

- Tablet 100 mg, 200 mg (scored)

How to Dose

- Titration up or down only necessary if not optimally efficacious at the standard starting dose of 200 mg once a day in the morning



Dosing Tips

- * For sleepiness, more may be more: higher doses (200–800 mg/day) may be better than lower doses (50–200 mg/day) in patients with daytime sleepiness in sleep disorders
- * For problems concentrating and fatigue, less may be more: lower doses (50–200 mg/day) may be paradoxically better than higher doses (200–800 mg/day) in some patients
- At high doses, may slightly induce its own metabolism, possibly by actions of inducing CYP450 3A4
- Dose may creep upward in some patients with long-term treatment due to autoinduction; drug holiday may restore efficacy at original dose

Overdose

- No fatalities; agitation, insomnia, increase in hemodynamic parameters
- Postmarketing experience includes CNS symptoms, such as restlessness, disorientation, confusion, excitation, and

hallucinations; digestive changes, such as nausea and diarrhea; and cardiovascular changes, such as tachycardia, bradycardia, hypertension, and chest pain

Long-Term Use

- Efficacy in reducing excessive sleepiness in sleep disorders has been demonstrated in 9- to 12-week trials
- Unpublished data show safety for up to 136 weeks
- The need for continued treatment should be reevaluated periodically

Habit Forming

- Schedule IV; may have some potential for abuse but unusual in clinical practice

How to Stop

- Taper not necessary; patients may have sleepiness on discontinuation

Pharmacokinetics

- Metabolized by the liver
- Excreted renally
- Elimination half-life 10–12 hours
- Inhibits CYP450 2C19 (and perhaps 2C9)

- Induces CYP450 3A4 (and slightly 1A2 and 2B6)

Drug Interactions

- May increase plasma levels of drugs metabolized by CYP450 2C19 (e.g., diazepam, phenytoin, propranolol)
- Modafinil may increase plasma levels of CYP450 2D6 substrates such as TCAs and SSRIs, perhaps requiring downward dose adjustments of these agents
- Modafinil may decrease plasma levels of CYP450 3A4 substrates such as ethinyl estradiol and triazolam
- Due to induction of CYP450 3A4, effectiveness of hormonal contraceptives may be reduced by modafinil, including 1 month after discontinuation
- Inducers or inhibitors of CYP450 3A4 may affect levels of modafinil (e.g., carbamazepine may lower modafinil plasma levels; fluvoxamine and fluoxetine may raise modafinil plasma levels)
- Modafinil may slightly reduce its own levels by autoinduction of CYP450 3A4
- Modafinil may increase clearance of drugs dependent on CYP450 1A2 and reduce their plasma levels
- Patients on modafinil and warfarin should have prothrombin times monitored
- Methylphenidate may delay absorption of modafinil by an hour

* However, coadministration with methylphenidate does not significantly change the pharmacokinetics of either modafinil or methylphenidate

* Coadministration with dextroamphetamine also does not significantly change the pharmacokinetics of either modafinil or dextroamphetamine

- Interaction studies with MAOIs have not been performed, but MAOIs can be given with modafinil by experts with cautious monitoring

Other Warnings/Precautions

- Patients with history of drug abuse should be monitored closely
- Modafinil may cause CNS effects similar to those caused by other CNS agents (e.g., changes in mood and, theoretically, activation of psychosis, mania, or suicidal ideation)
- Modafinil should be used in patients with sleep disorders that have been completely evaluated for narcolepsy, obstructive sleep apnea/hypopnea syndrome (OSAHS), and shift work sleep disorder
- In OSAHS patients for whom continuous positive airway pressure (CPAP) is the treatment of choice, a maximal effort to treat first with CPAP should be made prior to initiating modafinil, and then CPAP should be continued after initiating modafinil
- The effectiveness of hormonal contraceptives may be reduced when used with modafinil and for 1 month after discontinuation of

modafinil

- Modafinil is not a replacement for sleep

Do Not Use

- If patient has severe hypertension
- If patient has cardiac arrhythmias
- If there is a proven allergy to modafinil

Special Populations

Renal Impairment

- Use with caution; dose reduction is recommended

Hepatic Impairment

- Reduce dose by half in severely impaired patients

Cardiac Impairment

- Use with caution
- Not recommended for use in patients with a history of left ventricular hypertrophy, ischemic ECG changes, chest pain, arrhythmias, or recent myocardial infarction

Elderly

- Limited experience in patients over 65
- Clearance of modafinil may be reduced in elderly patients

Children and Adolescents

- Safety and efficacy not established under age 16
- Can be used cautiously by experts for children and adolescents

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
 - Controlled studies have not been conducted in pregnant women
 - Intrauterine growth restriction and spontaneous abortion have been reported with armodafinil and modafinil
 - In animal studies, developmental toxicity was observed at clinically relevant plasma exposures of armodafinil and modafinil
 - Use in women of childbearing potential requires weighing potential benefits to the mother against potential risks to the fetus
- * Generally, modafinil should be discontinued prior to anticipated pregnancies

Breast Feeding

- Unknown if modafinil is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Selective for areas of brain involved in sleep/wake promotion
- Less activating and less abuse potential than stimulants

Potential Disadvantages

- May not work as well as stimulants in some patients

Primary Target Symptoms

- Sleepiness
- Concentration
- Physical and mental fatigue

Pearls

- * Anecdotal usefulness for jet lag short-term (off label)
- * Modafinil is not a replacement for sleep
- * The treatment for sleep deprivation is sleep, not modafinil

- Controlled studies suggest modafinil improves attention in OSAHS, shift work sleep disorder, and ADHD (both children and adults), but controlled studies of attention have not been performed in major depressive disorder

* May be useful to treat fatigue in patients with depression as well as other disorders, such as multiple sclerosis, myotonic dystrophy, HIV/AIDS

- In depression, modafinil's actions on fatigue appear to be independent of actions (if any) on mood
- In depression, modafinil's actions on sleepiness also appear to be independent of actions (if any) on mood but may be linked to actions on fatigue or on global functioning
- Several controlled studies in depression show improvement in sleepiness or global functioning, especially for depressed patients with sleepiness and fatigue
- May be useful adjunct to mood stabilizers for bipolar depression
- May be useful in treating sleepiness associated with opioid analgesia, particularly in end-of-life management
- Subjective sensation associated with modafinil is usually one of normal wakefulness, not of stimulation, although jitteriness can rarely occur
- Anecdotally, some patients may experience wearing off of efficacy over time, especially for off-label uses, with restoration of efficacy

after a drug holiday; such wearing off is less likely with intermittent dosing

* Compared to stimulants, modafinil has a novel mechanism of action, novel therapeutic uses, and less abuse potential, but is often inaccurately classified as a stimulant

- Alpha 1 antagonists such as prazosin may block the therapeutic actions of modafinil
- The active R enantiomer of modafinil, called armodafinil, is also available

Suggested Reading

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